Vault Agent
by SportiQue

XRPL HACKATHON NYC 2026

Every data vault gets an agent that trades on its owner's terms.

80 days

to get one dataset from one institution to another.

35 of those days are the contract alone. The data was ready. The paperwork was not. Meanwhile data deals keep growing: AI training data alone goes from $3.2B to $16.3B by 2033.

Median wait on one major clinical data-sharing platform: YODA Project dashboard, July 2026 · Grand View Research

WHY IT IS SLOW

The bottleneck isn’t the data. It’s people coordinating and approving.

Step	Today, a person does it	Cost
Agree the terms	Contract negotiated by email	**35 of 80 days**
Consent	Every person approves every item, every round	**the slowest step**
Check quality	Reviewed at the end, after paying	**48.5% paid but unfit (synthetic)**
Pay	Batch approval, then card payout	**$0.88 fee on $20**

Cited: YODA 2026, card fee schedules · Synthetic: our market simulation

THE FIX: AN AGENT ON EACH SIDE

Each owner’s agent reads her vault and her rules, then trades on its own.

AGREEING TERMS
35 days → seconds

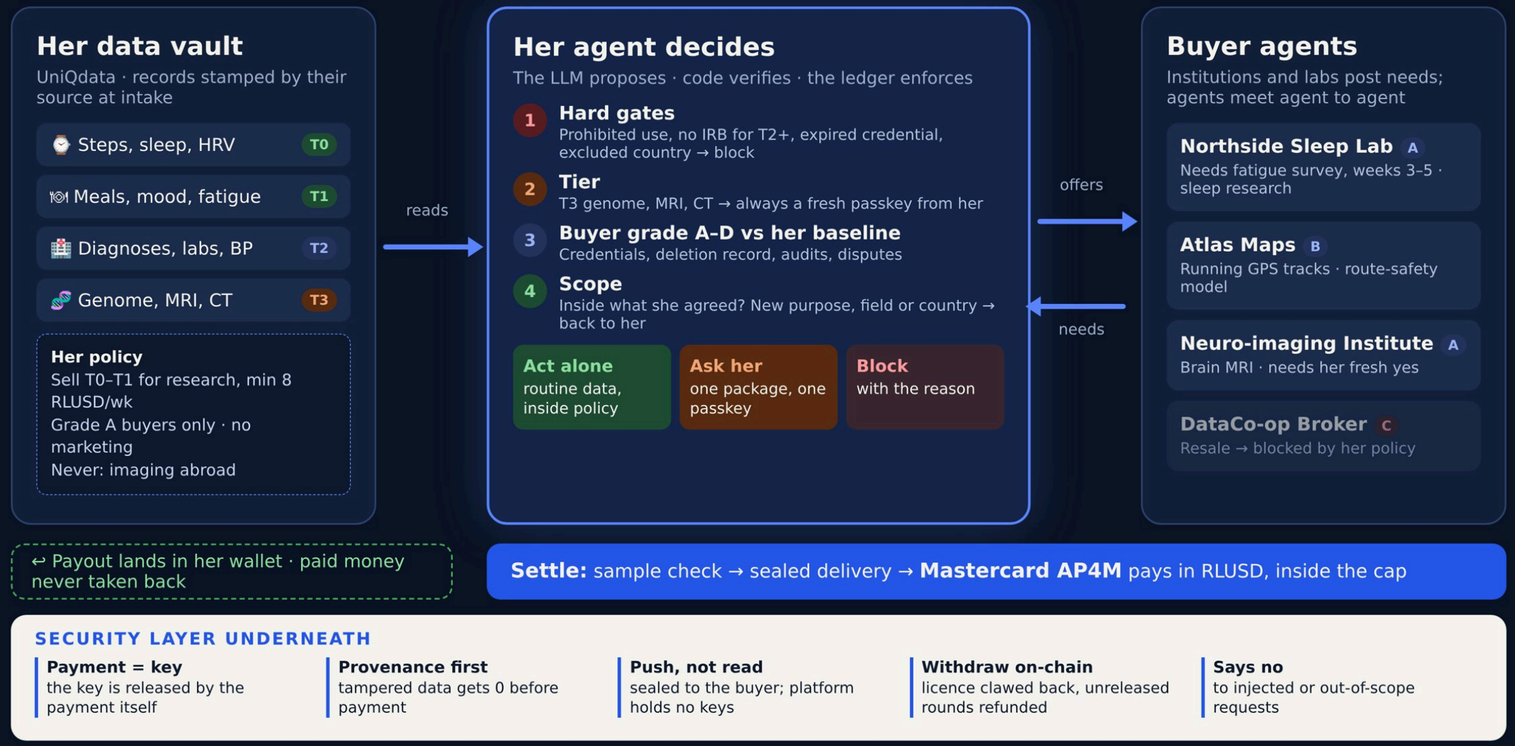
REQUEST → PAYMENT
12 h → < 1 min

PAID BUT UNFIT
48.5% → 19%

COST TO PAY $20
$0.88 → $0.0002

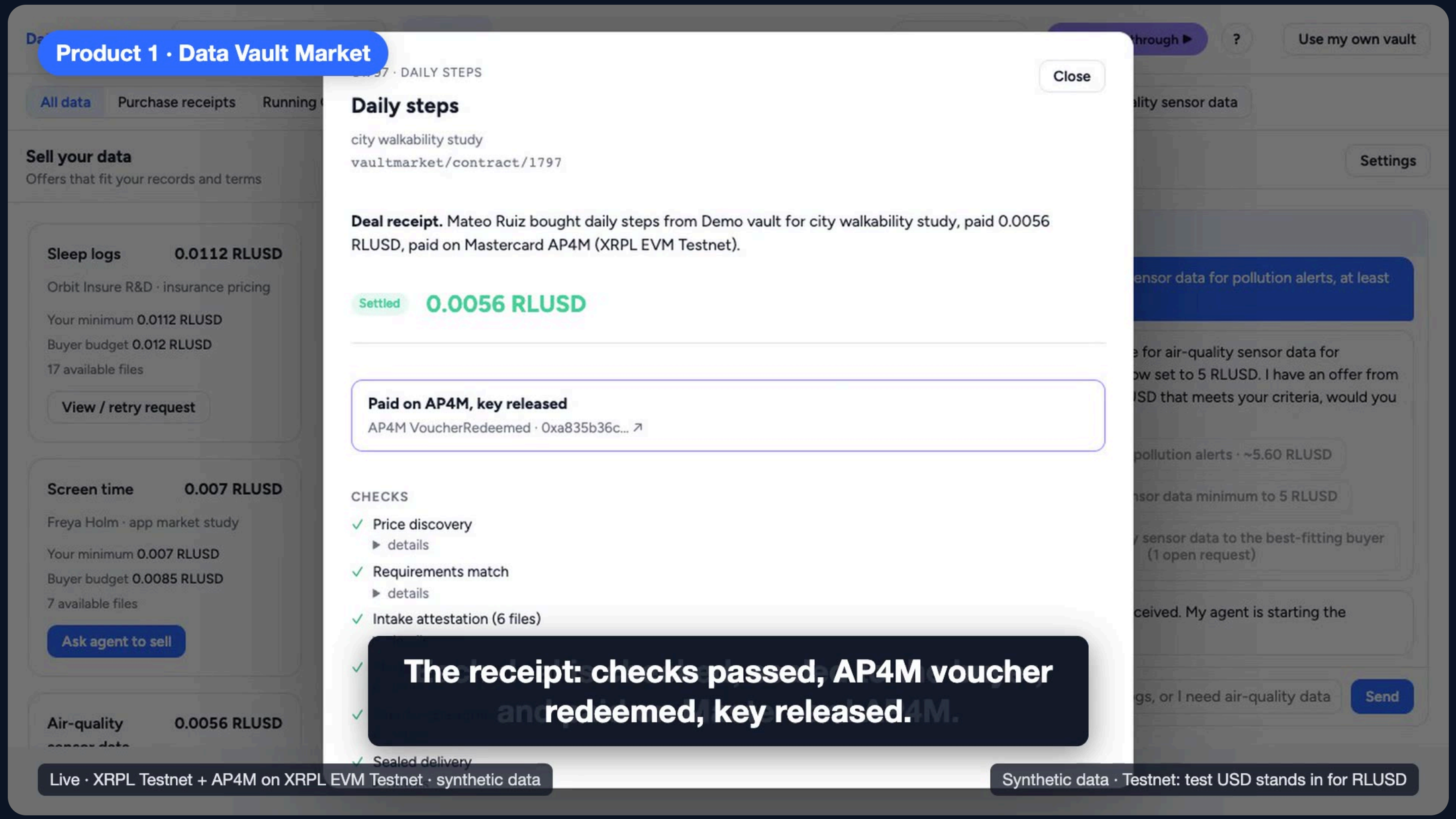
A personal data vault with its own agent

The agent reads her vault and her rules, decides on its own, finds the right buyer, negotiates and gets her paid. Sensitive data always waits for her.



Tiles, each its own source: YODA 2026 + our negotiation log · simulation with assumed response times, 50 people · market simulation (synthetic) · card vs XRPL fee

Vault agents negotiate and settle on their own.



PRODUCT 2 · UNIQDATA × UNIQLAB

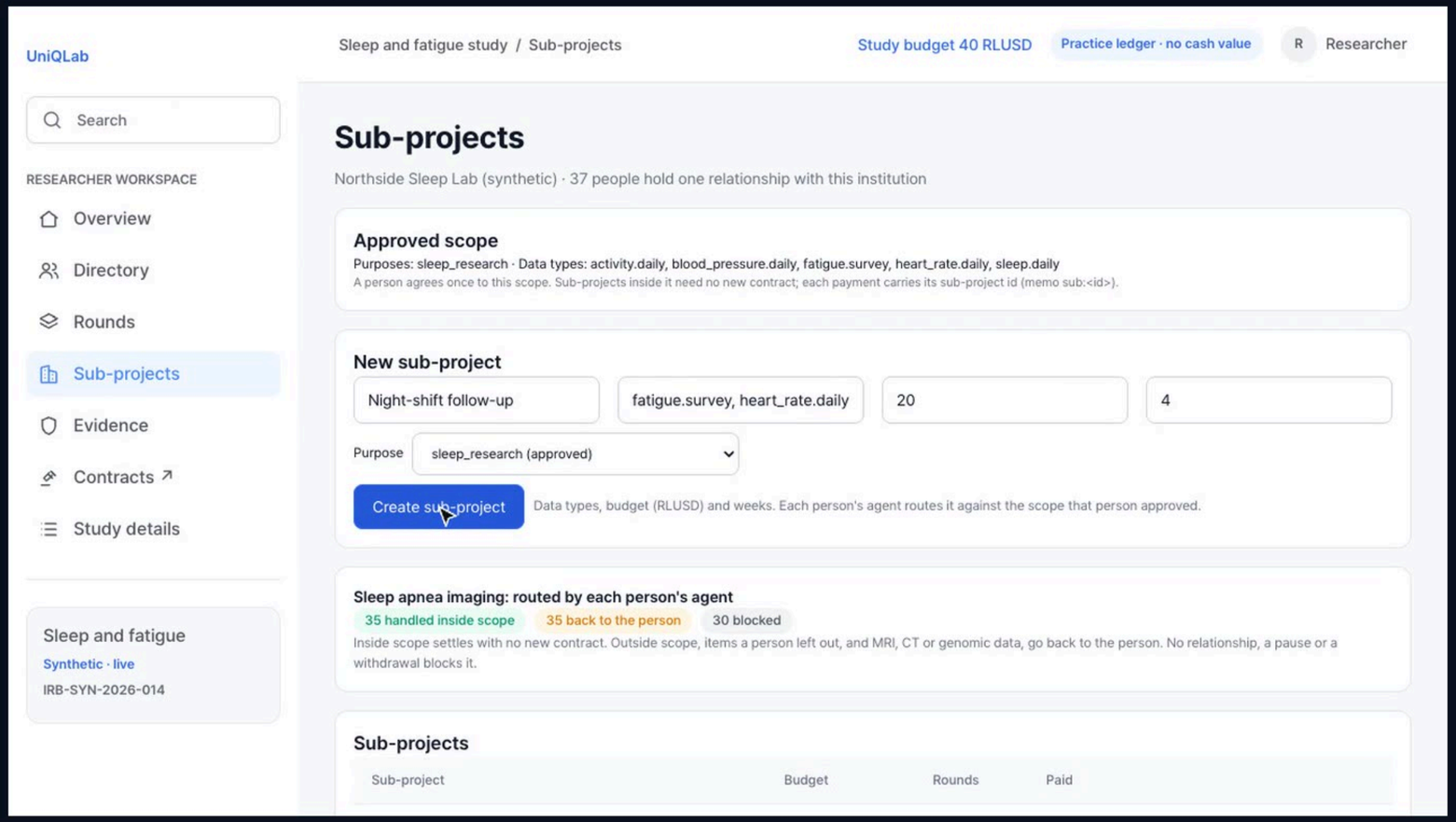
Consent by her policy. The lab's agent does the rest.

▶ Demo clip B · 62 s

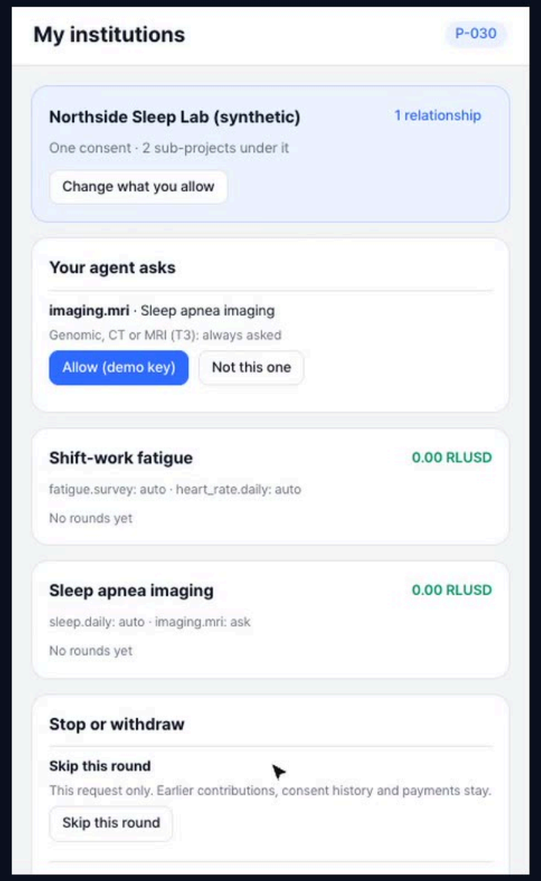
Vault Agent · UniQdata × UniQLab

one engine · both sides live · practice ledger, synthetic data

RESEARCHER · UNIQLAB CONSOLE



PARTICIPANT · UNIQDATA



T0 · everyday
steps, sleep, heart rate

T1 · lifestyle
meals, surveys

T2 · clinical
diagnoses · needs IRB

T3 · highly sensitive
MRI, CT, genome

MRI is T3 → stops for her fresh yes
Never covered by a standing grant · she taps Allow herself

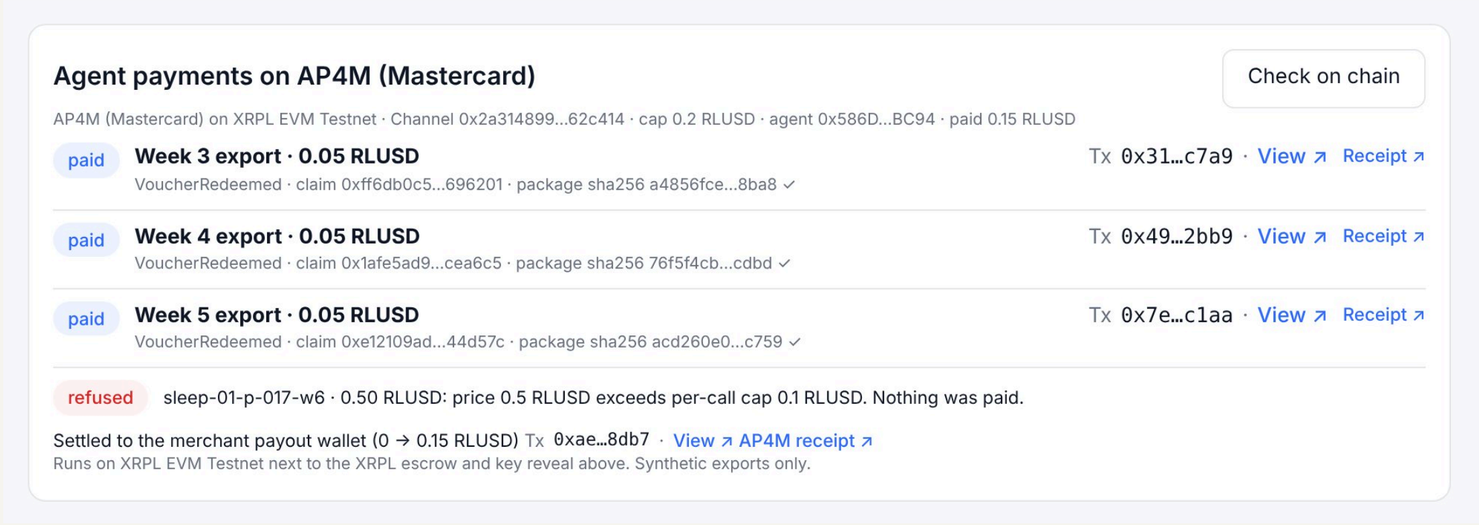
PROOF · LIVE ON MASTERCARD AP4M

One payment channel for both products.

3 paid · 1 refused

Weeks 3–5 paid 0.05 RLUSD each. Week 6 asked 0.50, over the 0.10 cap: refused, nothing paid. 3 of 3 receipts verified on-chain.

On XRPL: payment is the data key (escrow preimage), withdrawal claws back the licence, credentials gate who can buy.



Live AP4M run, Oct 25 2026, XRPL EVM Testnet · VoucherRedeemed 0x318d…, 0x4990…, 0x7e27… · payout wallet 0 → 0.15 RLUSD · Testnet: test USD stands in for RLUSD

WHY US

Others check the buyer’s budget. We also check the person’s terms.

Capability	Typical team	Vault Agent
Spend inside budget and caps	Yes	**Yes, via AP4M**
Pay only after a sample check	Some	**Yes**
A second approval: the person’s own terms	No	**Yes**
Sensitive data always needs a fresh yes	No	**Yes**
Payment releases the data key	No	**Yes**
Withdrawal is a ledger event	No	**Yes**

We lived this problem

SportiQue runs UniQdata and UniQLab. Consent friction with health data is our own daily problem, so the sensitivity layer comes from real study design, not a whiteboard.

Based on 17 public repositories reviewed at this hackathon (Oct 25). “No” means we did not find it there.

Vault Agent
by SportiQue

The owner sets the terms. Agents do the deal. AP4M pays.

NOW

Testnet: vault market, sensitivity tiers, AP4M payouts verified

3–6 MONTHS

Paid pilot with one longitudinal research team; ethics-reviewed consent

12 MONTHS

Mainnet RLUSD; sensitivity tier as an API for other data markets

Try it

Our ask: one research team for a paid pilot, and a path to mainnet RLUSD on AP4M.

Vault Agent by SportiQue · built on the AgentRail engine · synthetic data · XRPL Testnet and XRPL EVM Testnet · Testnet: test USD stands in for RLUSD

APPENDIX

For Q&A

Vault agent and eight scenarios · sensitivity tiers and how the agent decides · security layer · product walkthroughs · payout curve · rivals · legal basis · market · what is real today

A personal data vault with its own agent

The agent reads her vault and her rules, decides on its own, finds the right buyer, negotiates and gets her paid. Sensitive data always waits for her.

Her data vault

UniQdata · records stamped by their source at intake

- ⌚ Steps, sleep, HRV — T0
- 🍽 Meals, mood, fatigue — T1
- 🏥 Diagnoses, labs, BP — T2
- 🧬 Genome, MRI, CT — T3

Her policy
Sell T0–T1 for research, min 8 RLUSD/wk
Grade A buyers only · no marketing
Never: imaging abroad

reads

Her agent decides

The LLM proposes · code verifies · the ledger enforces

1. **Hard gates**
   Prohibited use, no IRB for T2+, expired credential, excluded country → block
2. **Tier**
   T3 genome, MRI, CT → always a fresh passkey from her
3. **Buyer grade A–D vs her baseline**
   Credentials, deletion record, audits, disputes
4. **Scope**
   Inside what she agreed? New purpose, field or country → back to her

Act alone	Ask her	Block
routine data, inside policy	one package, one passkey	with the reason

offers

needs

Buyer agents

Institutions and labs post needs; agents meet agent to agent

- **Northside Sleep Lab** A
  Needs fatigue survey, weeks 3–5 · sleep research
- **Atlas Maps** B
  Running GPS tracks · route-safety model
- **Neuro-imaging Institute** A
  Brain MRI · needs her fresh yes
- **DataCo-op Broker** C
  Resale → blocked by her policy

↩ Payout lands in her wallet · paid money never taken back

Settle: sample check → sealed delivery → **Mastercard AP4M** pays in RLUSD, inside the cap

SECURITY LAYER UNDERNEATH

- **Payment = key** — the key is released by the payment itself
- **Provenance first** — tampered data gets 0 before payment
- **Push, not read** — sealed to the buyer; platform holds no keys
- **Withdraw on-chain** — licence clawed back, unreleased rounds refunded
- **Says no** — to injected or out-of-scope requests

Eight scenarios, one loop: read the vault → decide → trade → pay out

Every scenario runs through the same agent, the same four checks, and the same Mastercard AP4M payment rail.

1 · B2B

Institution ↔ institution data deal

READ	Seller’s offer book and owner terms: min price, allowed uses
DECIDE	Buyer grade ≥ owner baseline; records not already owned
TRADE	Sample check, price from the last 20 ledger trades
PAY	AP4M in RLUSD, inside the buyer’s cap

Settled in seconds, no owner tap

2 · ROUTINE DATA

She sells sleep data automatically

READ	T0 sleep weeks + her policy: research only, ≥ 8 RLUSD/week
DECIDE	Inside her standing grant → act alone
TRADE	Sealed push to the lab, no read access
PAY	AP4M; her leg paid on to her wallet

Shared and paid while she sleeps

3 · SENSITIVE DATA

MRI, CT, genome wait for her

READ	Brain MRI request: grade A institute, grade C broker
DECIDE	T3 → fresh passkey for A; C held: “needs grade A+ and a controlled workspace”
TRADE	Only after her yes, inside a controlled workspace
PAY	AP4M after the key is released

Never automatic, never standing

4 · RESEARCHER AGENT

Selecting data across sub-projects

READ	Lab goal “back to 80% power” + gap map of missing weeks
DECIDE	Each person’s agent routes each sub-project: handled / back to her / blocked
TRADE	Requests exactly the missing person-weeks
PAY	Per verified week, through AP4M

No new contract per sub-project

5 · QUALITY

Tampered data refused before payment

READ	Source stamps signed when records entered her vault
DECIDE	Week 4 hash ≠ hospital’s signature → fail
TRADE	Stopped before any money moves
PAY	Nothing; seller grade drops

0 RLUSD for tampered data

6 · PAYOUT CURVE

Staying pays more, up front

READ	Her run of consecutive verified weeks
DECIDE	50% up front, +10 points a week, up to 90%; ×1.1/1.2/1.3 at 2/4/8 weeks
TRADE	The rest held 7 days (+14 if fraud suspected)
PAY	Paid by default, even after withdrawal

Paid money is never taken back

7 · WITHDRAWAL

She leaves; the ledger follows

READ	Her withdrawal, signed with a fresh passkey
DECIDE	Stop every next round for that study
TRADE	Research licence (MPT) clawed back; deletion requested and confirmed
PAY	Unreleased escrows refunded to the lab

What she earned stays hers

8 · CROSS-BORDER

A buyer in another country

READ	Buyer location vs her consents
DECIDE	Needs one-time cross-border consent per institution and country; “never abroad” types blocked
TRADE	Sealed copy pushed; no remote read access
PAY	AP4M as usual

Asked once, then automatic

Synthetic data · XRPL Testnet / XRPL EVM Testnet · Testnet: test USD stands in for RLUSD

OUR ANSWER · THE SENSITIVITY LAYER

Not all health data weighs the same. So the agent's freedom doesn't either.

Tier	Examples (46 columns catalogued)	What the person's agent may do
T0 · Daily	Steps, sleep, resting HR, HRV, SpO2	Act alone inside her policy (standing grant)
T1 · Lifestyle	Meals, caffeine, meds log, mood, fatigue	Act alone inside her policy
T2 · Clinical	Diagnoses, prescriptions, HbA1c, blood pressure	Only with a valid IRB credential; delegated only if she opts in
T3 · Highly sensitive	Genome, CT, MRI, mental-health diagnosis	Never alone, never standing: a fresh passkey every time

15-minute passkey grace

One passkey covers routine approvals for 15 minutes, bound to that consent package.

Always a fresh passkey

MRI, CT, genome · any new consent · withdrawal · changes to how she is paid

agentrail/catalog/columns.py · each column carries its tier, required GA4GH DUO codes, retention and deletion rule

HOW THE AGENT DECIDES

Every request passes four checks, in order. Each answer comes with a reason.

1 · HARD GATES

Prohibited use, no IRB for T2+, expired credential, excluded country, a type she marked “never”

→ blocked. No setting widens it.

2 · TIER

T3 (genome, CT, MRI)?

→ always a fresh yes from her, never automatic.

3 · BUYER GRADE

Buyer trust A–D (credentials, deletion record, audits, disputes) vs her baseline per data type

→ grade ≥ baseline: act.

4 · SCOPE

Inside what she already agreed? New purpose, field, institution type or country?

→ inside: act. New: one package back to her.

Same MRI request	Grade A lab → asks her for a fresh yes. Grade C lab → held without asking: *“MRI needs grade A+ and a controlled workspace, buyer is C.”*

The LLM proposes, code verifies, the ledger enforces. The LLM can only narrow a request, never widen it (tested).

THE SECURITY LAYER UNDERNEATH

Money, keys and consent are tied together on the ledger.

Payment is the key

The data key K is the escrow's preimage. The transaction that pays is the one that unlocks.

Stamped at intake

Hospitals and devices sign records as they enter her DB. A tampered week fails before payment: 0 RLUSD.

Push, not read

Sealed end to end to the buyer's key. Nobody gets read access to her store; the platform holds no keys or funds.

One order, one record

Consent hash, bundle hash and escrow are bound into one trade record anyone can check.

Withdrawal on-chain

Her research licence (MPT) is clawed back; unreleased rounds are refunded. Delivered rounds can't be recalled, and she is told first.

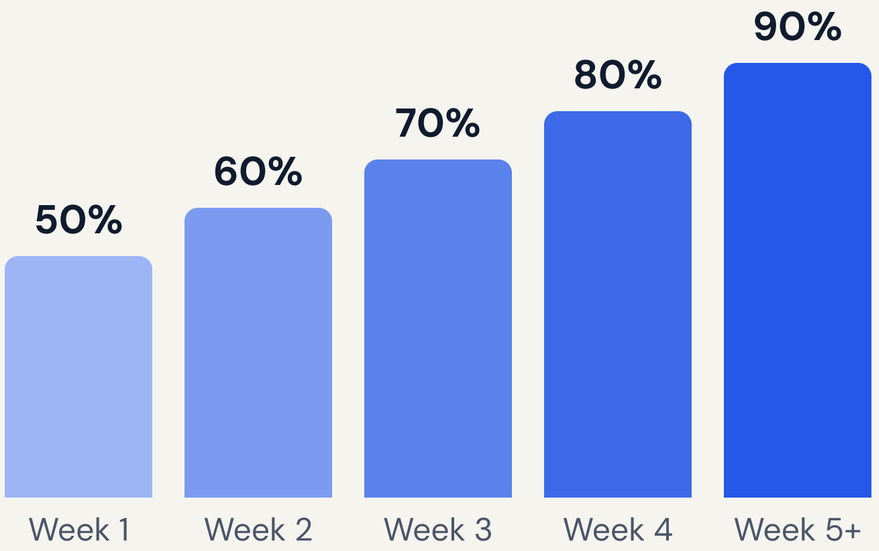
An agent that says no

Expired ethics credential, out-of-scope field, "ignore consent" hidden in a record: refused, 0 orders.

25 attack-replay tests in tests/test_security.py · each XRPL primitive verified on Testnet, including its expected rejection

FAIR PAY FOR STAYING

The longer she contributes, the more she gets paid up front.



Paid at key reveal: 50% of a verified week, +10 points per consecutive week, up to 90%.

The rest: held 7 days (+14 if fraud is suspected), then paid by default, even after she withdraws.

Streak bonus: weekly rate ×1.1 / ×1.2 / ×1.3 after 2 / 4 / 8 weeks.

Never clawed back: money already paid stays hers. A tampered week gets 0 before payment.

agentrail/payout_schedule.py · FDA 2018: payment should accrue as the study progresses · Testnet: test USD stands in for RLUSD

WHO WE ARE

SportiQue builds research on personal health databases: one person, many studies.

UniQdata

Participant app. People gather their wearable, clinic and survey records in one personal DB and decide which studies get what.

UniQLab

Researcher console. Institutions design studies, request exactly the data they need, and pay per contribution.

One-to-many

One relationship with an institution, many projects under it. No new contract each time.

WE HIT THE SAME WALL, HARDER

In health, every deal touches a person. Consent friction kept participation low.

18

Consent is tedious

Every item, every study, every week by hand (3 items × 6 weeks, illustrative)

?

Hard to judge

Consent forms are legal text; most postings don't fit the person scrolling them

15–40%

People drop out

Replacing one costs ≈ $19.5k; remote wearable studies keep people a median 5.5 days

$0

People get nothing

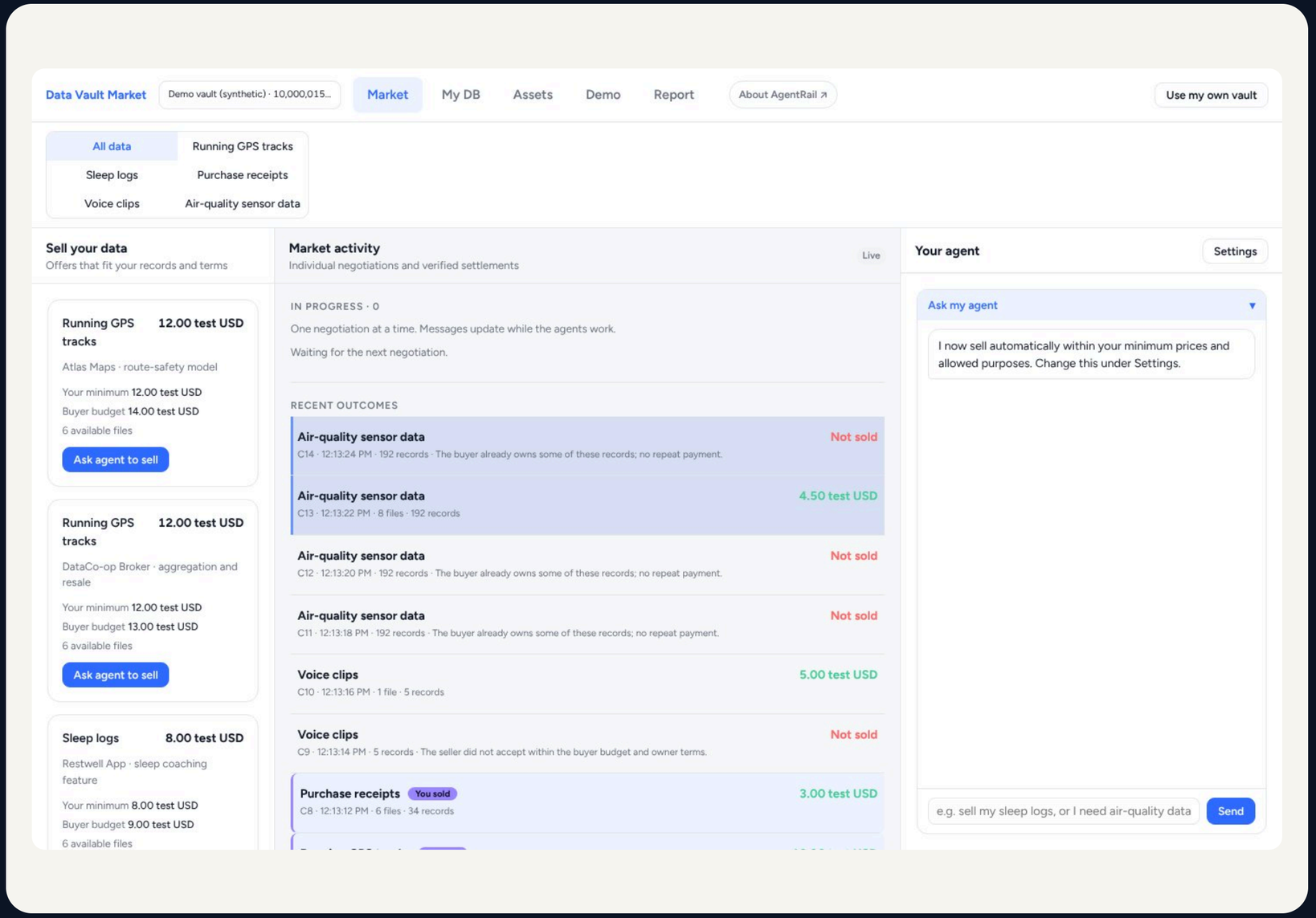
23andMe's DNA data sold for $305M in bankruptcy; customers received nothing

Sources: Clinical Leader; Medidata dropout model; TTAM acquisition reports · 18 approvals is an illustration, not a SportiQue measurement

APPENDIX · DATA VAULT MARKET

Tell your agent what to sell or buy. It does the rest.

- Offers that fit your records and terms appear on the left
- Every negotiation and settlement shows live in the middle
- “Not sold” always comes with a reason
- Your agent sells only above your minimum, for allowed purposes



Data Vault Market (live) · demo vault, synthetic data · Testnet: test USD stands in for RLUSD

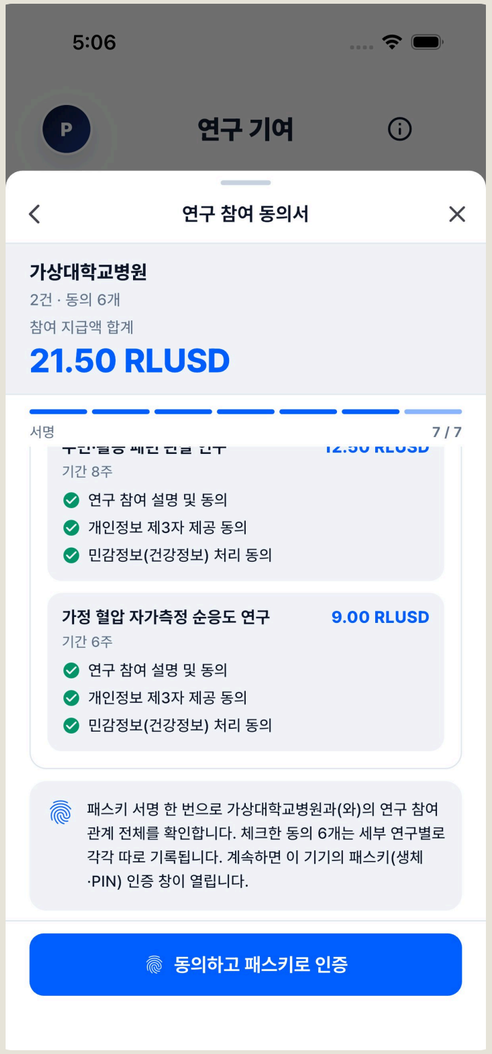
APPENDIX · UNIQDATA × UNIQLAB

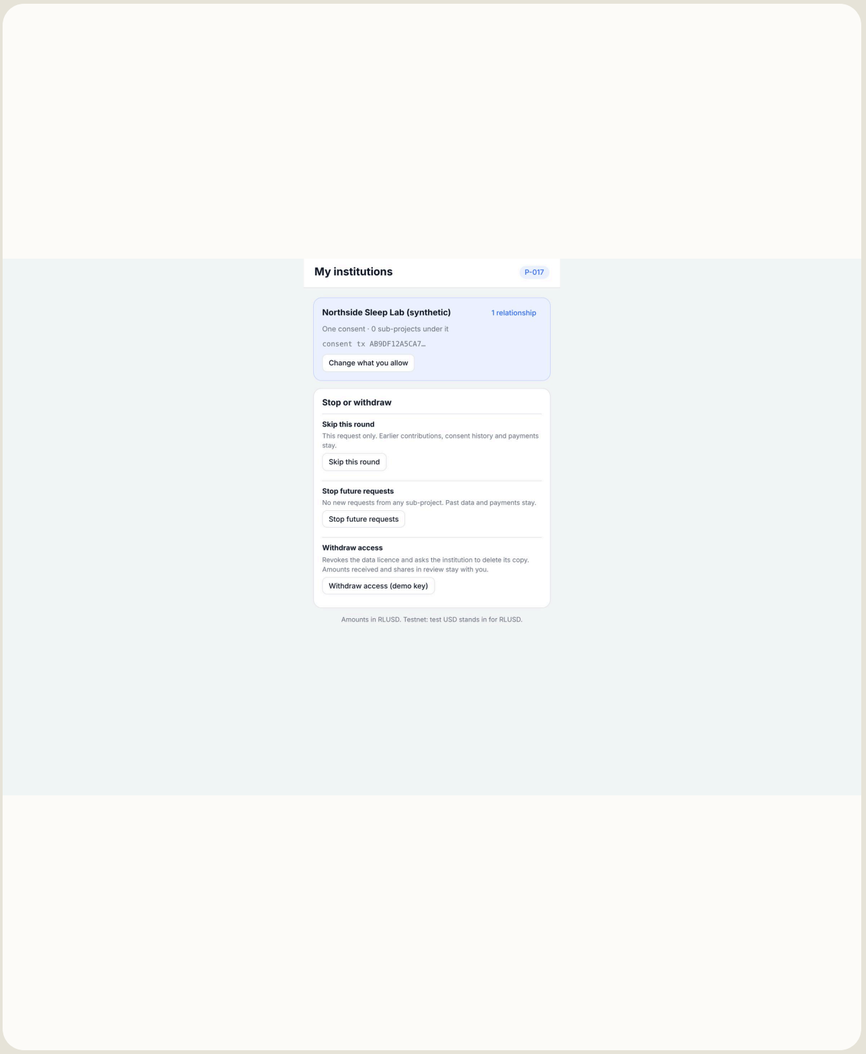
The person signs once. Her agent handles the rest.

- Clauses must be read to the end before the box unlocks
- One passkey for the institution; each consent recorded per study
- Routine requests then go through her agent; anything new returns to her

11,600 → 233

person's actions: column-by-column approvals vs passkey taps (simulation, 50 people, assumed response times)





UniQdata Flutter app (demo build) and participant page · synthetic studies · Testnet: test USD stands in for RLUSD

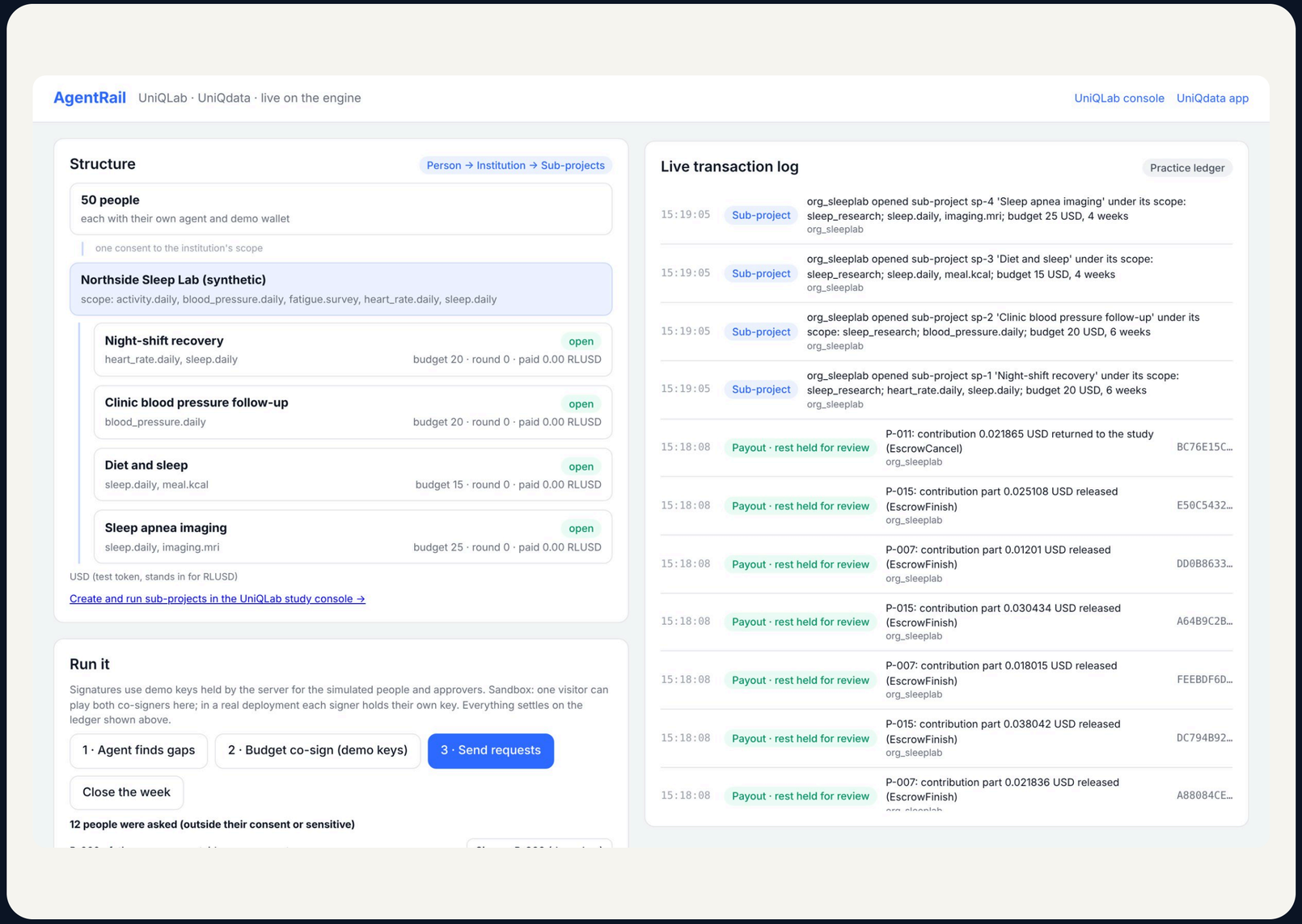
APPENDIX · UNIQDATA × UNIQLAB

Our contract structure, on one screen.

Institution ↔ institution
Agent-to-agent data deals in the vault market

Institution ↔ person
One relationship, one consent to the institution's scope

Many sub-projects under it
No new contract each time; every payment carries its sub-project id



Vault Agent (formerly AgentRail) live page: 50 people · Northside Sleep Lab (synthetic) · 4 sub-projects · live transaction log · Testnet: test USD stands in for RLUSD

APPENDIX · UNIQDATA × UNIQLAB

Every new sub-project is routed by each person's agent.

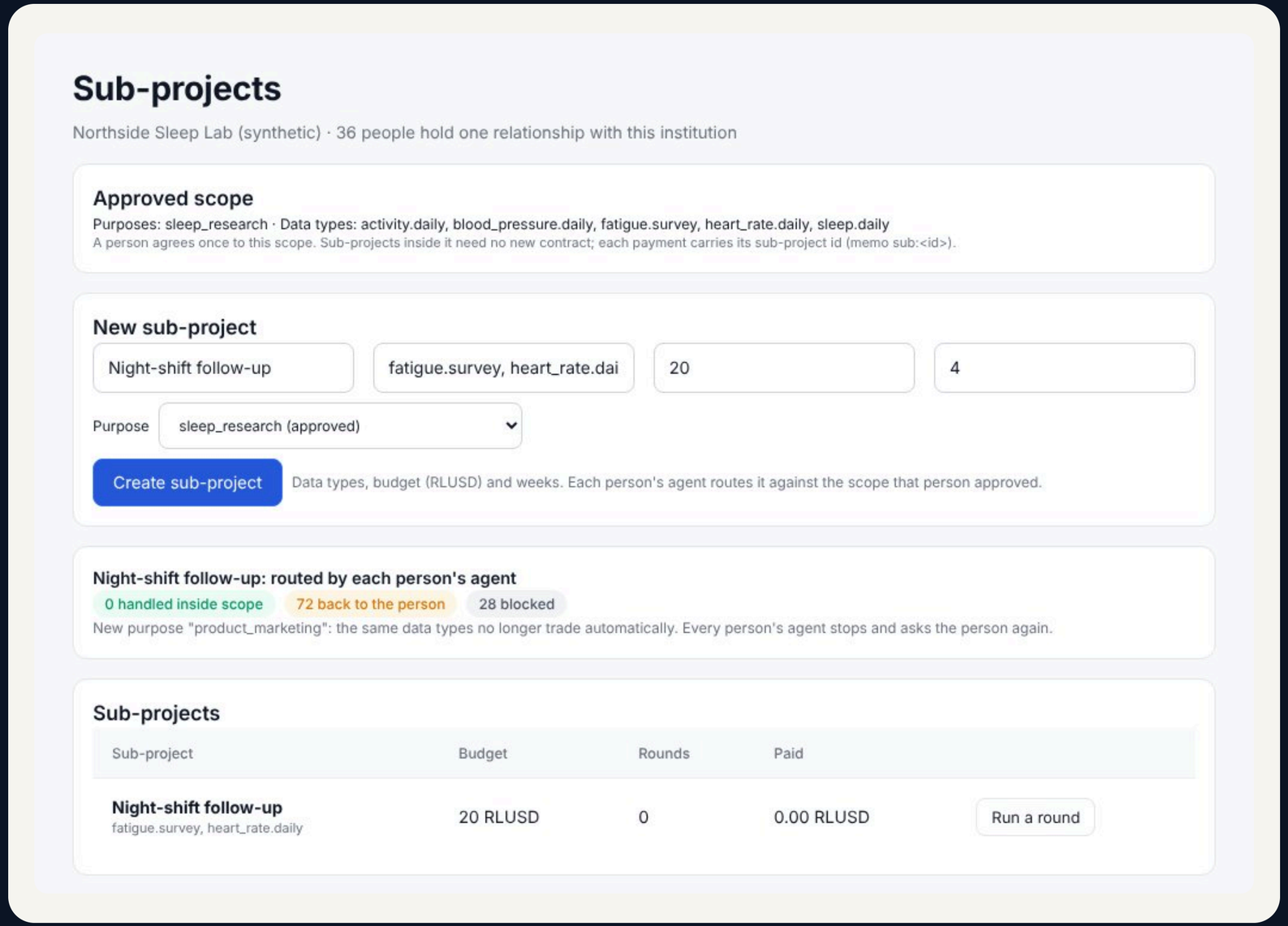
The lab opens "Night-shift follow-up". Every person's agent checks each request against what that person approved, at once.

inside scope → handled

new purpose → back to her

not allowed → blocked

Here the purpose changed to product marketing, so 0 requests trade automatically: 72 go back to their person, 28 are blocked.

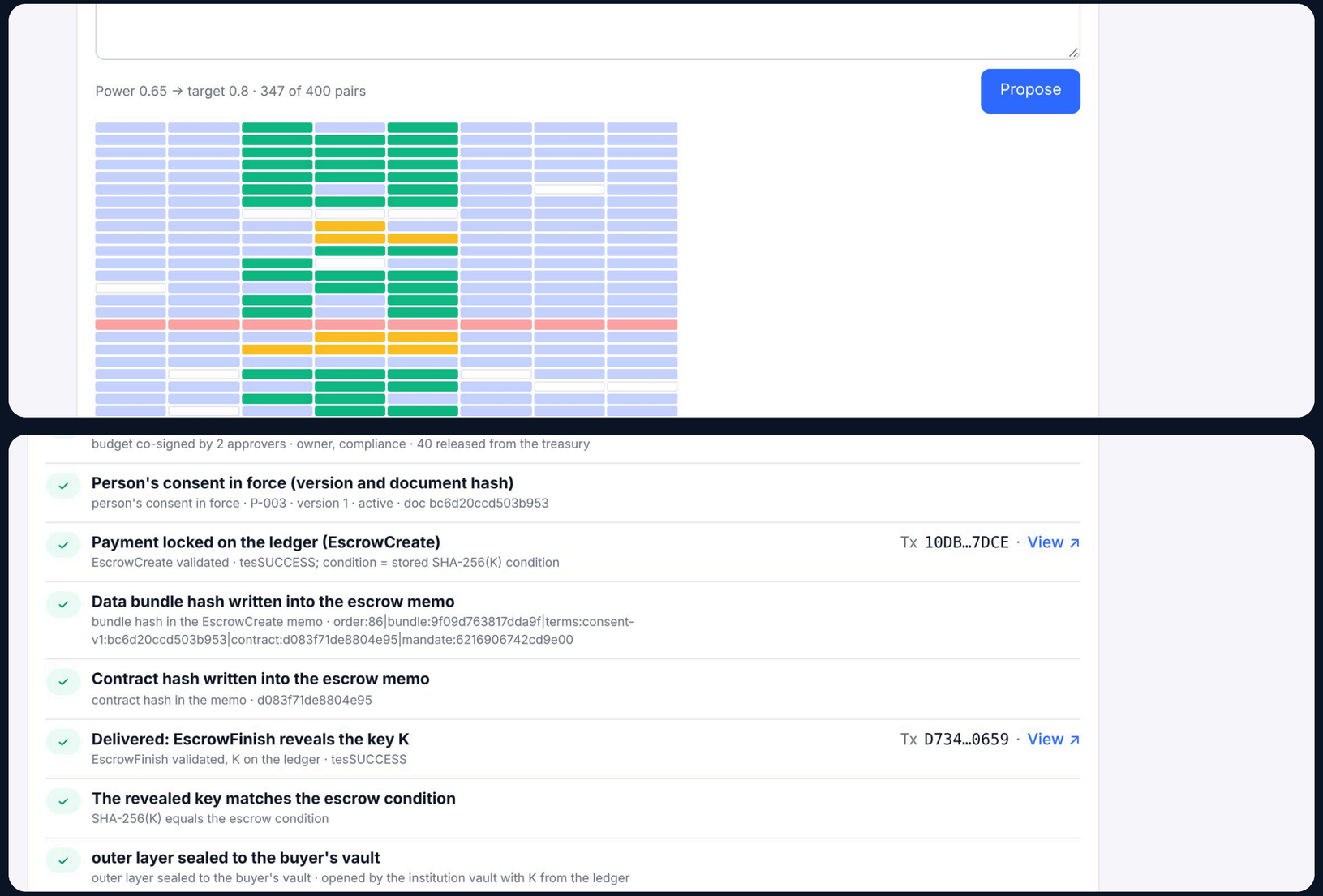


UniQLab researcher console · Sub-projects · synthetic cohort

APPENDIX · UNIQDATA × UNIQLAB

The researcher's agent does the hand work.

By hand today	Agent
Spot gaps at analysis	Gap map from each contribution
Hand-filter data	Sample check before paying
Chase approvals	Budget co-signed by 2
Batch payouts	Paid per verified week



UniQLab console: Rounds (top) and Evidence of one trade (bottom) · mock ledger run · synthetic data

WHAT CHANGES

More sharing for people. Better-fit data for researchers.

For the person’s agent

- Shares routine data on its own, so people share more often
- More contributions means more income, paid per round in RLUSD
- Sovereignty kept: sensitive data and new uses always come back to them
- Withdraw any time; payments already received stay theirs

For the researcher’s agent

- Reads the protocol and finds exactly which data is missing
- Compares offers agent to agent, not person to person
- Checks a sample before paying; unfit data is dropped first
- Gets cleaner, consented data, faster and safer

WHY IT IS DIFFERENT

Others check the buyer's budget. We also check the person's terms.

Capability	Typical agent-payment team	Vault Agent
Spend inside budget, sellers, caps	Yes	Yes, via Mastercard AP4M
Pay only after delivery is checked	Some	Yes, sample check first
A second approval: the person's own terms	No	Yes, per institution, purpose, field, period
Sensitive data always needs a fresh yes	No	Yes (MRI, CT, genome)
Payment releases the key to the data	No (money only)	Yes
Withdrawal is a ledger event	No	Yes, licence clawed back
Agent turns money down for the person	No	Yes

Based on 17 public repositories reviewed at this hackathon (Oct 25). "No" means we did not find it there.

KOREA · UNITED STATES · MONGOLIA

We turned legal requirements into product rules.

The law asks	Product rule	Basis
Separate consent for health data	Sensitive, third-party and cross-border consents signed separately, each hashed	KR PIPA Art. 22–23 · 45 CFR 46.116
No use beyond the consent	Code re-checks recipient, purpose, fields, period every round and before key release	KR PIPA Art. 17 · FTC HBNR
Viewing from abroad is a transfer	One-time cross-border consent per institution and country, shown in plain words; push, never read access	KR PIPA Art. 28-8 · MN PDPL Art. 14.1
Withdraw any time, no penalty	Skip, pause, withdraw; withdrawal never affects pay or priority	KR PIPA Art. 37 · Bioethics Act Art. 16 · 46.116(b)(8)
IRB decides, not the platform	Payment releases only with the IRB's on-ledger credential	Bioethics Act Art. 15 · 45 CFR 46.112
No custody, no personal data on-chain	Institutions sign their own payments; ledger holds hashes only	FinCEN 2019-G001 · PIPC blockchain guideline 2026

Not legal advice; designed to the requirements, not "compliant". Mongolia A/90 server rule reviewed before launch. Demo: synthetic data, Testnet.

MARKET

Start with the hardest data. Then lighten the layer for everything else.

BEACHHEAD

≈ $2.5B

Person-level longitudinal health research (SAM). A dropout costs ≈ $19.5k; a missing person-week ≈ $20.

EXPAND

$298B+

Data agents will buy: brokers, real-world evidence, AI training data, robot sensor data.

YEAR 3 (ESTIMATE)

$11.4M

300 studies × 500 people × 26 weeks; 10% take rate + $12k per study per year.

Directional; definitions vary by source · market-deep-dive.md, market-potential.md

WHAT IS REAL TODAY

Clear about what runs, what is simulated, and what is next.

Runs on-chain

- AP4M payouts: 3 paid, 1 refused, receipts verified
- Key-reveal escrow, MPT licence clawback, credentials, two-signer multisig budget, each with its expected rejection
- RLUSD-style token escrow

Simulated

- All people, studies and records are synthetic
- Market of 50 accounts (80 of 109 paid deals by agents)
- Approval counts use assumed response times

Next

- Real UniQdata participants under ethics review
- Mainnet RLUSD
- Written legal opinions (Mongolia A/90 first)